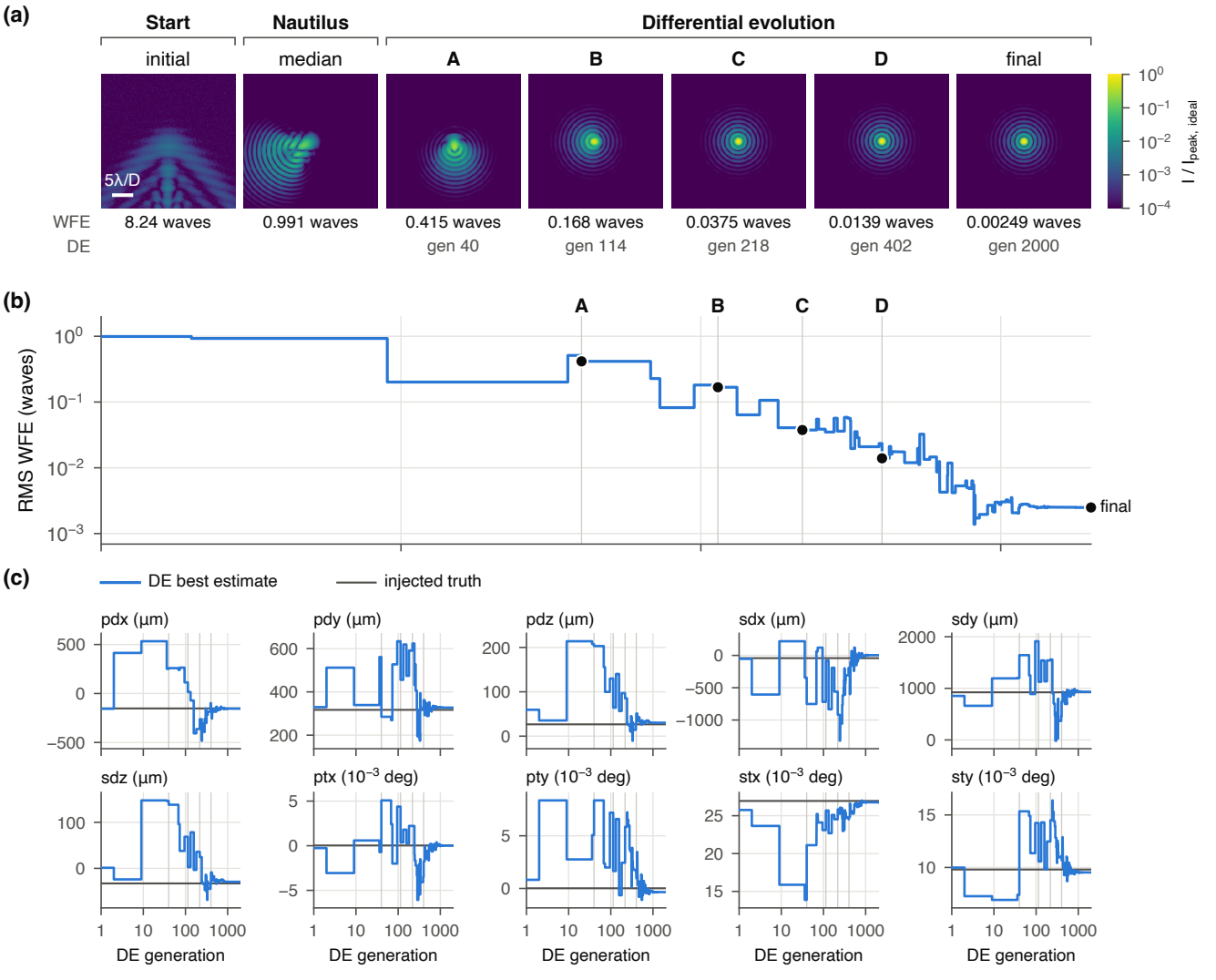

(a)
Start
Nautilus
Differential evolution
initial
median
A
B
C
D
final
5λ/D
WFE
DE
8.24 waves
0.991 waves
0.415 waves
0.168 waves
0.0375 waves
0.0139 waves
0.00249 waves
gen 40
gen 114
gen 218
gen 402
gen 2000
I / I peak, ideal
(b)
RMS WFE (waves)
final
(c)
DE best estimate
injected truth
pdx (μm)
pdy (μm)
pdz (μm)
sdx (μm)
sdy (μm)
sdz (μm)
ptx (10−3 deg)
pty (10−3 deg)
stx (10−3 deg)
sty (10−3 deg)
DE generation